

and is usually referred as the point O. The equation $x = a \sec \theta$, $y = b \tan \theta$ are called the parametric eqns of hyperbola where θ is called the parameter.

Q. Find the eqn of chord joining points θ and ϕ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Let the points θ and ϕ be represented by
 $\theta \rightarrow (a \sec \theta, b \tan \theta)$ $\phi \rightarrow (a \sec \phi, b \tan \phi)$

Then the chord joining these two points are

$$\frac{y - b \tan \theta}{x - a \sec \theta} = \frac{b \tan \phi - b \tan \theta}{a \sec \theta - a \sec \phi}$$

$$\frac{y - b \tan \theta}{b \tan \phi - b \tan \theta} = \frac{x - a \sec \theta}{a \sec \theta - a \sec \phi}$$

$$\frac{y/b - \tan \theta}{\tan \phi - \tan \theta} = \frac{x/a - \sec \theta}{\sec \phi - \sec \theta}$$

$$y/b (\sec \theta - \sec \phi) - \tan \theta (\sec \phi - \sec \theta) = x/a (\tan \phi - \tan \theta) - \sec \theta (\tan \phi - \tan \theta)$$

$$x/a (\tan \phi - \tan \theta) - y/b (\sec \phi - \sec \theta) = \sec \theta (\tan \phi - \tan \theta) - \tan \theta (\sec \phi - \sec \theta)$$

$$x/a (\tan \phi - \tan \theta) - y/b (\sec \phi - \sec \theta) = \sec \theta \tan \phi - \sec \theta \tan \theta - \tan \theta \sec \phi + \tan \theta \sec \theta$$

$$= \frac{1}{\cos \theta} \frac{\sin \phi}{\cos \phi} - \frac{\sin \theta}{\cos \theta} \frac{1}{\cos \phi}$$

$$= \frac{\sin \phi - \sin \theta}{\cos \theta \cos \phi}$$

$$\therefore \frac{x}{a} \left(\frac{\sin \phi - \sin \theta}{\cos \phi \cos \theta} \right) - \frac{y}{b} \left(\frac{1}{\cos \phi} - \frac{1}{\cos \theta} \right) = \frac{\sin \phi - \sin \theta}{\cos \theta \cos \phi}$$

$$\frac{x}{a} (\sin \phi \cos \theta - \sin \theta \cos \phi) - \frac{y}{b} (\cos \theta - \cos \phi) = \sin \phi - \sin \theta$$

$$\frac{x}{a} \sin(\phi - \theta) - \frac{y}{b} (\cos \theta - \cos \phi) = \sin \phi - \sin \theta = -(\sin \theta - \sin \phi)$$

$$\frac{x}{a} \sin(\theta - \phi) + \frac{y}{b} (\cos \theta - \cos \phi) = \sin \theta - \sin \phi$$

$$\frac{x}{a} 2 \sin\left(\frac{\theta - \phi}{2}\right) \cos\left(\frac{\theta + \phi}{2}\right) + \frac{y}{b} 2 \sin\left(\frac{\theta + \phi}{2}\right) \sin\left(\frac{\theta - \phi}{2}\right)$$

$$= 2 \cos\left(\frac{\theta + \phi}{2}\right) \sin\left(\frac{\theta - \phi}{2}\right)$$

$$\therefore \boxed{\frac{x}{a} \cos\left(\frac{\theta + \phi}{2}\right) - \frac{y}{b} \sin\left(\frac{\theta + \phi}{2}\right) = \cos\left(\frac{\theta + \phi}{2}\right)}$$

Q. Find the eqn of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

We know that eqn of chord at $(a \sec \theta, b \tan \theta)$ and $(a \sec \phi, b \tan \phi)$ is

$$\frac{x}{a} \cos\left(\frac{\theta-\phi}{2}\right) - \frac{y}{b} \sin\left(\frac{\theta+\phi}{2}\right) = e \cos\left(\frac{\theta+\phi}{2}\right)$$

This chord will become tangent at θ if $\theta = \phi$

$$\therefore \frac{x}{a} \cos \theta - \frac{y}{b} \sin \theta = \cos \theta$$

$$\frac{x}{a} - \frac{y}{b} \sin \theta = \cos \theta \Leftrightarrow \boxed{\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1}$$

Q. Find eqn of normal at point θ on hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

The eqn of tangent at θ is

$$\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$$

$$\Rightarrow y = \frac{bx}{a} - \frac{y}{b} \tan \theta = \frac{x}{a} \sec \theta - 1$$

$$\Rightarrow y = \frac{b}{a} \frac{\sec \theta}{\tan \theta} x - \frac{b}{\tan \theta}$$

$$= \frac{b}{a} \operatorname{cosec} \theta - b \cot \theta$$

$\therefore$ Slope of tangent is $\frac{b}{a} \operatorname{cosec} \theta$

$\therefore$ Slope of normal = $-1 / \frac{b}{a} \operatorname{cosec} \theta$

$$= -\frac{a}{b} \sin \theta$$

∴ Eqn of normal with slope $-\frac{a}{b} \tan \theta$ and passing through $(a \sec \theta, b \tan \theta)$ is

$$y - b \tan \theta = -\frac{a}{b} \tan \theta (x - a \sec \theta)$$

$$yb - b^2 \tan \theta = -xa \tan \theta + a^2 \tan \theta$$

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$$ax \tan \theta + yb = \tan \theta (a^2 + b^2)$$

$$\boxed{\frac{ax}{\sec \theta} + \frac{yb}{\tan \theta} = a^2 + b^2}$$

Q. Find the condition for the line $lx + my + n = 0$ to touch the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

The eqn of tangent to hyperbola is

$$a^2 \sec^2 \theta - b^2 \tan^2 \theta = 1$$

$$\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$$

Given $lx + my + n = 0$

$$\frac{\sec \theta}{a} / l = \frac{-\tan \theta}{b} / m = 1 / -n$$

$$\frac{l \sec \theta}{a} = \frac{1}{-n} \Rightarrow \sec \theta = \frac{-a}{ln}$$

$$\tan \theta = \frac{bm}{an}$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\frac{a^2 l^2}{n^2} - \frac{b^2 m^2}{n^2} = 1$$

$$\Rightarrow \underline{\underline{a^2 l^2 - b^2 m^2 = n^2}}$$

Q. Find the condition for the line ~~for~~ $lx + my + n = 0$ to be a normal to hyperbola.

The eqn of normal is

$$\frac{ax}{\sec \theta} + \frac{yb}{\tan \theta} = a^2 + b^2 \quad \text{--- (1)}$$

$$lx + my = -n \quad \text{--- (2)}$$

(1) and (2) represent the same line.

$$\frac{a/\sec \theta}{l} = \frac{b/\tan \theta}{m} = \frac{a^2 + b^2}{-n}$$

$$\frac{la}{l \sec \theta} = \frac{mb}{m \tan \theta} = \frac{a^2 + b^2}{-n}$$

$$\sec \theta = \frac{-na}{a^2 + b^2}$$

$$\tan \theta = \frac{-mbn}{a^2 + b^2}$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\frac{n^2 a^2}{l^2} - \frac{b^2 n^2}{m^2} = (a^2 + b^2)^2$$

$$\underline{\underline{\frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}}}$$

4/12/19

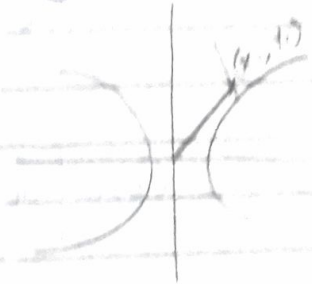
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(Saathi)

Q. 5.7 The locus of feet of L^s from the centre to the normals of the hyperbola is $(x^2+y^2)^2(a^2y^2-b^2x^2)=(a^2+b^2)x^2y^2$

The eqn of normal to hyperbola at $(a \sec \theta, b \tan \theta)$ is

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$



$$\text{i.e. } ax \sec \theta + by = (a^2 + b^2) \tan \theta \quad \text{--- (1)}$$

Let (x_1, y_1) be foot of L^s drawn from centre to eqn(1) i.e. the normal

$$(1) \Rightarrow ax_1 \sec \theta + by_1 = (a^2 + b^2) \tan \theta \quad \text{--- (2)}$$

Slope of L^s is given by

$$\frac{y_1 - 0}{x_1 - 0} = \frac{y_1}{x_1}$$

$(0,0)$
 (x_1, y_1)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope of (1) is } -\frac{a \sec \theta}{b}$$

$$\therefore \text{Slope of (1)} \times \text{Slope of } L^s = -1$$

$$-\frac{a \sec \theta}{b} \times \frac{y_1}{x_1} = -1$$

$$\sec \theta = \frac{x_1 b}{y_1 a} \quad \text{--- (3)}$$

Sub (3) in (2)

$$ax_1 \sin \theta + by_1 = (a^2 + b^2) \frac{\sin \theta}{\cos \theta}$$

$$\Rightarrow \cos \theta \left[\frac{ax_1}{y_1} + b \right] = (a^2 + b^2) \frac{\sin \theta}{ay_1}$$

$$\Rightarrow \cancel{\cos \theta} = \frac{(a^2 + b^2)b x_1}{a y_1} \cdot \cancel{\frac{y_1^2}{x_1^2 + y_1^2}}$$

$$\Rightarrow \cos \theta \left(\frac{x_1^2}{y_1} + y_1 \right) = (a^2 + b^2) \frac{x_1}{ay_1}$$

$$\Rightarrow a \cos \theta (x_1^2 + y_1^2) = (a^2 + b^2) x_1 y_1$$

$$\Rightarrow \cos \theta = \frac{x_1 (a^2 + b^2)}{a (x_1^2 + y_1^2)}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\Rightarrow \left(\frac{bx_1}{ay_1} \right)^2 + \left(\frac{x_1 (a^2 + b^2)}{a (x_1^2 + y_1^2)} \right)^2 = 1$$

$$\Rightarrow \frac{b^2 x_1^2 (x_1^2 + y_1^2)^2 + x_1^2 (a^2 + b^2)^2 y_1^2}{y_1^2 (x_1^2 + y_1^2)^2} = a^2$$

Date _____

Saathi

$$(a^2 + b^2)^2 \underline{x_1^2 y_1^2} = (x_1^2 + y_1^2)^2 (a^2 y_1^2 - b^2 x_1^2)$$

Hence proved.